



THE ENVIRONMENTAL PARADOX Of Tech Manufacturing

Electronics powers our world, but at what cost? Massive energy consumption, hazardous chemicals, and waste plague PCB and IC manufacturing. Can tech go green before it is too late?



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The manufacturing of printed circuit boards (PCBs) and integrated circuits (ICs) stands at the core of our tech-driven world but brings significant environmental concerns. In 2023 alone, Taiwan Semiconductor Manufacturing Company (TSMC), a leading industry player, used a staggering 25,000 gigawatt hours of energy and approximately 101 million cubic metres of water. These figures highlight the immense resource consumption typical in semiconductor manufacturing, which involves millions of silicon wafers and hazardous chemicals. Given the current environmental challenges, how sustainable are these practices?

PCB manufacturing mirrors these environmental concerns. The process heavily relies on chemicals such as ferric chloride and copper chloride, particularly during the etching phase. Moreover, the primary material used is Flame Retardant 4 (FR4), a type of epoxy resin derived from petroleum, underscoring the reliance on fossil

fuels. The procedure is notably wasteful, discarding about 70% of copper through subtractive methods. This not only wastes valuable resources but also significantly harms the environment. Have you considered the scale of waste generated by such processes?

Disposal poses another major issue. At the end of their life cycle, vast amounts of metals, plastics, and other materials from electronic devices often end up in landfills, leading to potential soil and water contamination. Some are incinerated to recover energy, which, unfortunately, contributes to air pollution.

However, there is a silver lining as the industry pursues improvements through operational efficiencies and sustainability goals. Techniques such as reducing manufacturing temperatures, streamlining process steps, and optimising chemical use are gaining traction. Research into new materials that are potentially more sustainable or less harmful is also underway.